

SIDDHARTHA SHAKYA

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PROFILE SUMMARY

First-Class Honors Computer Science graduate specializing in end-to-end Data Science and Machine Learning life cycle. Hands-on experience across the MLOps lifecycle, CI/CD automation, and cloud infrastructure (AWS/GCP). Equipped with practical internship experience and seeking to deliver scalable data solutions in an entry-level Data Science role.

TECHNICAL SKILLS

Languages	Python, SQL, C/C++
Databases	PostgreSQL, MySQL, MongoDB
Cloud	AWS, GCP
AI & ML	PyTorch, Scikit-learn, MLflow, Pandas, NumPy
Data Engineering	Databricks, dbt, Apache Spark & Airflow
Data Analytics	Google Sheets, Google Colab, Plotly, Seaborn, Jupyter Notebook
DevOps	Docker, GitHub Actions, GitHub, Git

WORK EXPERIENCE

- AI Intern** | *Mach24 Orbitals* **Oct 2025 - Feb 2026**
- Built and deployed a Streamlit dashboard on GCP processing Landsat-8 satellite imagery across 5+ years of time series data; enabled real-time visualization of vegetation indices, land surface temperature, and urbanization patterns for environmental monitoring. Dashboard processes 1,000+ imagery tiles monthly with 200ms average load time.
 - Evaluated 3+ regression models (Random Forest, XGBoost, SVM) for Above Ground Biomass (AGB) estimation using Sentinel-2 imagery via Google Earth Engine API across 450+ test plots. XGBoost achieved 23% lower RMSE than Random Forest baseline (RMSE: 14.2 vs. 18.5 Mg/ha), identified as production candidate for scaled deployment.
- Graduate Teaching Assistant** | *Herald College Kathmandu* **Aug 2024 - Sep 2025**
- Led 8+ workshop sessions on parallel computing and performance optimization, instructing 50+ students across programming paradigms and GPU computing fundamentals; direct feedback integration improved student assessment scores by 15%.
 - Conducted hands-on labs and tutorials on Big Data technologies covering Hadoop ecosystem (HDFS, MapReduce), distributed processing frameworks (Spark), and real-time streaming pipelines; designed and graded practical assignments for 3+ cohorts (120 students total).

PROJECTS

- EduFin Investigation** | 2026 [Project Link](#)
- Executed a comprehensive risk analysis on a 15,000+ student loan portfolio to investigate a Rs. 12 Crore discrepancy using advanced SQL and data pipelines.
 - Uncovered key default drivers by segmenting geographic, demographic, and institutional risk factors, delivering actionable insights through interactive executive dashboards.
 - **Tech Stack:** Databricks, PySpark SQL, Python, Delta Tables, Data Visualization (Dashboards)
- Geospatial Environmental Monitoring Dashboard** | 2025 [Project Link](#)
- Conducted time-series geospatial analysis on USGS satellite data to quantify environmental trends in vegetation, water bodies, and urbanization from 2022-2024.
 - **Tech Stack:** Python, Pandas, GeoPandas, Streamlit, Folium, Plotly, Docker, GCP

EDUCATIONAL BACKGROUND

- PG Diploma in Data Science** **Feb 2026 – present**
Manipal Academy of Higher Education
- BSc (Hons) Computer Science** **Feb 2022 – Jun 2024**
University of Wolverhampton
Honors: **First Class**

CERTIFICATIONS

SQL Bootcamp	Udemy	Certificate
Complete MLOps Bootcamp	Udemy	Certificate
PySpark: Building Data Pipelines	LinkedIn Learning	Certificate